



GUIDEM ACADEMY • CYBERSECURITY TRAINING PROGRAM

EHVAPT

Ethical Hacking: Vulnerability Assessment & Penetration Testing

Complete Course Outline — Full Curriculum, Lab Schedule & Certification Path

5

Modules

78

Lessons

20

Hands-on Labs

20

CTF Challenges

COURSE OVERVIEW

What This Course Delivers

EHVAPT is a hands-on offensive security program that takes learners from core InfoSec and vulnerability assessment concepts through full-scope penetration testing, infrastructure foothold, web application exploitation, and introductory exploit development. Every module pairs lecture with a live lab against a dedicated attack range, closing with a practical GMCPT certification exam.



Format

Instructor-led lectures + guided labs + independent CTF challenges



Prerequisites

Basic networking (TCP/IP), Linux command line, general IT literacy



Certification

GMCPT — GuideM Certified Penetration Tester (practical exam)



Audience

Aspiring pentesters, SOC analysts, VAPT & red team practitioners



5

Core Modules



78

Total Lessons



20

Practical Labs



20

CTF Challenges

Duration guidance: 5 intensive training days (8 hrs/day) or a 4-week part-time track (2 sessions/week)

InfoSec Fundamentals & Vulnerability Assessment

Building the foundation: security principles, risk, and systematic vulnerability discovery

- CIA Triad & core security principles
- Threat landscape & threat actor profiles
- Risk management & asset classification
- Vulnerability Assessment (VA) methodology
- Scanning with Nessus & OpenVAS
- CVSS scoring & vulnerability prioritization
- Compliance touchpoints (ISO 27001, NIST)
- VA reporting & remediation tracking



HANDS-ON LAB

VA Sweep

Run an authenticated vulnerability scan across a mixed Windows/Linux subnet using OpenVAS, score findings with CVSS v3.1, and produce a prioritized remediation report.



Penetration Testing & Real-World Use Cases

From methodology to mission: scoping, reconnaissance, and engagement execution

- PTES / OWASP testing methodology phases
- Rules of Engagement & scoping documents
- Passive OSINT & footprinting techniques
- Active recon: Nmap, Shodan, subdomain enum
- Legal & ethical considerations, authorization
- Real-world case studies & breach post-mortems
- Reporting standards for client deliverables
- Red team vs. blue team engagement models



HANDS-ON LAB

Recon Chain

Execute full passive-to-active reconnaissance against a sanctioned target range, chaining OSINT, subdomain discovery, and service enumeration into a scoped attack surface map.

Establishing Foothold & Moving Deeper into Infrastructure

Gaining access, escalating privilege, and moving laterally through a network

- Initial access & credential attacks
- Brute force & password spraying
- Windows privilege escalation techniques
- Linux privilege escalation techniques
- Docker & container escape / misconfig abuse
- Lateral movement & pivoting
- Persistence mechanisms
- Post-exploitation cleanup & OPSEC



HANDS-ON LAB

Foothold to Domain

Compromise an initial Windows Server host via a weak service, escalate privilege using scheduled-task and AlwaysInstallElevated abuse, then pivot to a second segmented host.

Web & Web Application Technologies Fundamentals

Understanding the modern web stack and exploiting the OWASP Top 10



- HTTP/HTTPS protocol deep dive
- Sessions, cookies, JWT & authentication flows
- OWASP Top 10: SQLi, XSS, IDOR, SSRF
- REST & GraphQL API testing
- Burp Suite & OWASP ZAP proficiency
- Automated scanning with sqlmap
- Business logic & access control flaws
- Bug bounty workflow & disclosure practice



HANDS-ON LAB

DVWA Breakout

Exploit SQL injection with sqlmap, chain an IDOR into unauthorized data access, and bypass a broken authentication flow inside a deliberately vulnerable web application.

Exploit Development Basics & PowerShell Fundamentals

Introductory binary exploitation and offensive PowerShell scripting

- Memory layout & buffer overflow basics
- Fuzzing & crash triage fundamentals
- Building a simple stack overflow exploit
- PowerShell fundamentals for offense
- AMSI & basic AV/EDR evasion concepts
- PowerShell-based post-exploitation scripting
- Introductory malware analysis concepts
- Capstone: chained exploit walkthrough



HANDS-ON LAB

Stack Smash & Script

Build a working proof-of-concept buffer overflow exploit against a vulnerable Windows binary, then author a PowerShell post-exploitation script for data collection.

DELIVERY SCHEDULE

5-Day Intensive Lab Schedule



Day 1

Foundations & Recon

InfoSec fundamentals, VA methodology, OSINT & active recon



Day 2

Vulnerability Assessment

Nessus/OpenVAS scanning, CVSS scoring, VA reporting lab



Day 3

Foothold & Escalation

Credential attacks, Windows/Linux privesc, lateral movement



Day 4

Web Application Testing

OWASP Top 10, Burp Suite, sqlmap, API/GraphQL testing



Day 5

Exploit Dev & Capstone

Buffer overflow basics, PowerShell scripting, GMCPT exam

Each day runs 08:00–17:00: 3 hrs lecture, 4 hrs guided lab, 1 hr capstone/CTF challenge review.

Tools & Technology Stack



Reconnaissance

Nmap, theHarvester, Shodan, Amass, Recon-ng



Vulnerability Assessment

Nessus, OpenVAS, Nikto, Qualys



Exploitation & Foothold

Metasploit, Hydra, Mimikatz, sqlmap



Web Application Testing

Burp Suite, OWASP ZAP, Postman, ffuf



Post-Exploitation

PowerSploit, BloodHound, Impacket, Cobalt Strike-style TTPs



Reporting & Documentation

Dradis, CVSS calculators, Markdown/DOCX report templates

PRACTICAL ASSESSMENT

Hands-On Labs & CTF Challenges

20 guided labs build core technique; 20 independent CTF-style challenges validate mastery without a walkthrough.

Module	Guided Labs	CTF Challenges	Focus Area
Module 1 — VA Fundamentals	4 labs	4 CTFs	Scan tuning, CVSS scoring, report drafting
Module 2 — Pentest Methodology	4 labs	4 CTFs	OSINT chaining, scoping exercises, recon-to-report
Module 3 — Foothold & Infrastructure	4 labs	4 CTFs	Privesc chains, lateral movement, pivoting
Module 4 — Web Application Testing	4 labs	4 CTFs	OWASP Top 10 exploitation, API abuse
Module 5 — Exploit Dev & PowerShell	4 labs	4 CTFs	Buffer overflow build, offensive scripting

Final capstone: a chained 5-stage attack scenario (Recon → Brute Force → SQLi → DoS → Detection/Mitigation), mirroring authorized red-vs-blue demo format.

Assessment & GMCPT Certification

1



Module Quizzes

Short knowledge checks after each of the 5 modules to reinforce core concepts.

2



Lab Completion

All 20 guided labs must be completed and submitted with evidence (screenshots/logs).

3



CTF Scoring

20 independent CTF challenges scored on a points system; minimum threshold required.

4



GMCPT Practical Exam

A timed, scenario-based exam: compromise a target network end-to-end and submit a professional pentest report.



Passing learners earn the GMCPT (GuideM Certified Penetration Tester) credential — recognized proof of practical, end-to-end offensive security capability.

COURSE OUTCOMES

What You Walk Away With



Systematic vulnerability assessment & CVSS-based reporting



Real-world foothold, privilege escalation & lateral movement skills



Practical web application exploitation across the OWASP Top 10



Introductory exploit development & offensive PowerShell scripting



GMCPT certification — a portfolio-ready, practical credential



NEXT STEP

From EHVAPT to the Field

Graduates are positioned for SOC Analyst, Junior Pentester, and VAPT Associate roles — or to continue toward Security+, eJPT, and OSCP.